Problem 05: BioDiversity Scan

Time limit: 1 second

Global temperature raise is one of the most significant problems in current era. Recently, the world leaders announced Carbon Credits to minimize carbon emissions. One way to monitor Carbon Credits is to check the quality of eco system in a particular region. This quality can be determined by the presence of some specific insects in that region. The most common insects are ladybug, butterfly, and fireflies.

Dr. Wang has developed an AI based automated eco-friendly insects' detection system. This system is installed in a remote area and send the number of total detected insects each day. A minimum of M number of each N insect (ladybug, butterfly, fireflies etc.) is required to maintain a healthy eco system. Let's assume Dr. Wang system detected a total of T insects in one day. Write a program that can determine the number of ways we can have a total count of T for N insects, assuming each insect's category has a minimum of M number. For example, if $N = 3$, $T = 34$ and $M = 10$ then the possible combinations are as follow:

14 10 10
13 11 10
13 10 11
12 11 11
12 10 12
…….
and so on.

Input

In the first line of the input there will be a single positive integer K followed by K lines each containing a single test case. Each test case contains three positive integers denoting $\mathbf{0} \leq N \leq \mathbf{10}, \mathbf{0} \leq T \leq \mathbf{200}$ and $\mathbf{0} \leq M \leq \mathbf{10}$ respectively, separated by a single space.

Output

For each input, print the value of total number of possible combinations. Otherwise print 0 if not possible.

Sample:

S.No	Sample Input	Sample Output
1	2 3 34 10 2 40 15	15 11
2	1 30 3 15	0